

Notes on Continuum Mechanics (an excerpt)

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1.5 Algebraic Operations with Tensors

1.5.1 Dyadic

The *tensor product*, made up of two vectors $\vec{\mathbf{v}}$ and $\vec{\mathbf{u}}$, becomes a dyad, which is a particular case of a second-order tensor. The dyad is represented by:

$$\vec{\mathbf{u}}\vec{\mathbf{v}} \equiv \vec{\mathbf{u}} \otimes \vec{\mathbf{v}} = \mathbf{A} \quad (1.72)$$

where the operator $\otimes$ denotes the *tensor product*. Then, we define a *dyadic* as a linear combination of dyads. Furthermore, as we will see later, any tensor can be represented by means of a linear combination of dyads, (see Holzapfel (2000)).

The tensor product has the following properties:

$$1. \quad (\vec{\mathbf{u}} \otimes \vec{\mathbf{v}}) \cdot \vec{\mathbf{x}} = \vec{\mathbf{u}}(\vec{\mathbf{v}} \cdot \vec{\mathbf{x}}) \equiv \vec{\mathbf{u}} \otimes (\vec{\mathbf{v}} \cdot \vec{\mathbf{x}}) \quad (1.73)$$

$$2. \quad \vec{\mathbf{u}} \otimes (\alpha \vec{\mathbf{v}} + \beta \vec{\mathbf{w}}) = \alpha \vec{\mathbf{u}} \otimes \vec{\mathbf{v}} + \beta \vec{\mathbf{u}} \otimes \vec{\mathbf{w}} \quad (1.74)$$

$$3. \quad (\alpha \vec{\mathbf{v}} \otimes \vec{\mathbf{u}} + \beta \vec{\mathbf{w}} \otimes \vec{\mathbf{r}}) \cdot \vec{\mathbf{x}} = \alpha (\vec{\mathbf{v}} \otimes \vec{\mathbf{u}}) \cdot \vec{\mathbf{x}} + \beta (\vec{\mathbf{w}} \otimes \vec{\mathbf{r}}) \cdot \vec{\mathbf{x}} \\ = \alpha [\vec{\mathbf{v}} \otimes (\vec{\mathbf{u}} \cdot \vec{\mathbf{x}})] + \beta [\vec{\mathbf{w}} \otimes (\vec{\mathbf{r}} \cdot \vec{\mathbf{x}})] \quad (1.75)$$

where α and β are scalars. By definition, the dyad does not contain the commutative property, *i.e.*, $\vec{\mathbf{u}} \otimes \vec{\mathbf{v}} \neq \vec{\mathbf{v}} \otimes \vec{\mathbf{u}}$.

The equation (1.72) can also be expressed in the Cartesian system as:

$$\begin{aligned} \mathbf{A} = \vec{\mathbf{u}} \otimes \vec{\mathbf{v}} &= (u_i \hat{\mathbf{e}}_i) \otimes (v_j \hat{\mathbf{e}}_j) \\ &= u_i v_j (\hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j) \quad (i, j = 1, 2, 3) \\ &= A_{ij} (\hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j) \end{aligned} \quad (1.76)$$

$$\boxed{\underbrace{\mathbf{A}}_{\text{Tensor}} = \underbrace{A_{ij}}_{\text{components}} \underbrace{\hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j}_{\text{basis}}} \quad (i, j = 1, 2, 3) \quad (1.77)$$

In this textbook, the components of a second-order tensor can be represented in different ways, namely:

$$\begin{aligned} \mathbf{A} &= \vec{\mathbf{u}} \otimes \vec{\mathbf{v}} \\ &\quad \downarrow \\ &\quad \text{components} \\ &\quad \downarrow \\ (\mathbf{A})_{ij} &= (\vec{\mathbf{u}} \otimes \vec{\mathbf{v}})_{ij} = u_i v_j = A_{ij} \end{aligned} \quad (1.78)$$

These components are explicitly expressed in matrix form as: